

HAT-P-36b

R-0900

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_240131 · created 2026-07-29T03:26:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460340.9845 +/- 0.0054 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.082 +/- 0.04
Transit depth (Rp/Rs)^2	0.67 +/- 0.65 [%]
Orbital Inclination (inc)	86.2 +/- 2.77
Airmass coefficient 1 (a1)	1.224 +/- 0.026
Airmass coefficient 2 (a2)	-0.19 +/- 0.13
Scatter in the residuals of the lightcurve fit is	2.11 %
Best Comparison Star	None
Optimal Aperture	3.24
Optimal Annulus	8.65
Transit Duration (day)	0.0882 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.082 $\pm$ 0.04 vs 0.126 (deviation 1.01 σ)
Duration fit vs published	0.0882 d vs 0.0925 d (deviation 0.49 σ)
Mid-time O-C	12.5 min
Depth SNR	1.9
Residual scatter	2.11 %

Light curve

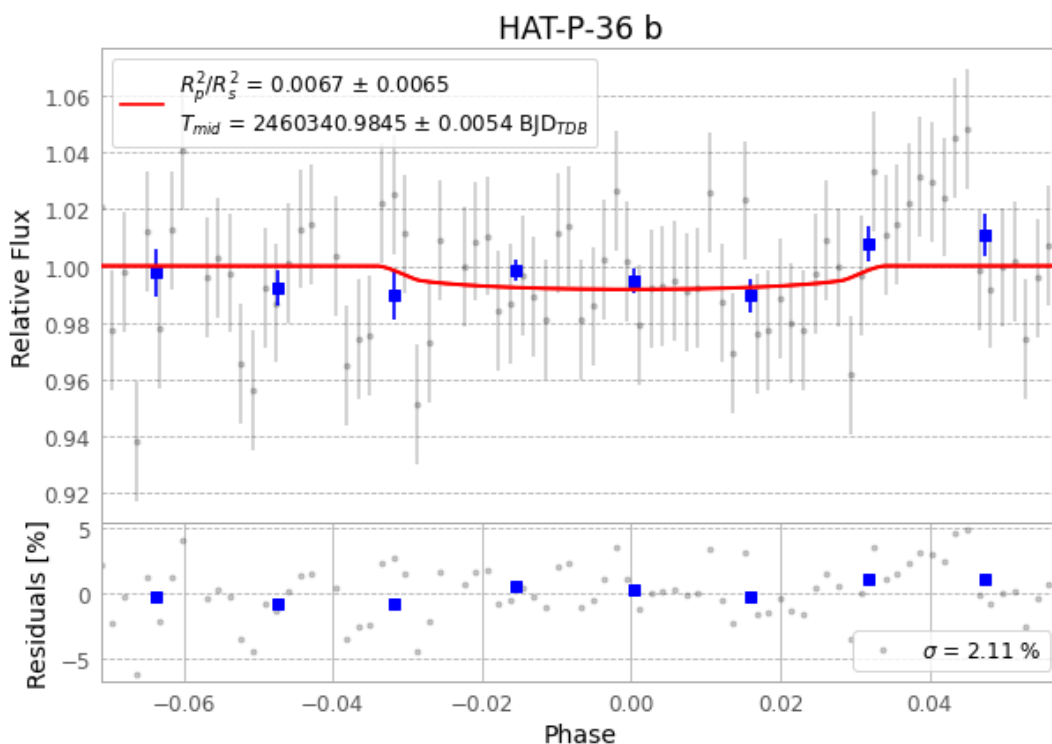


Figure 1 — FinalLightCurve_HAT-P-36 b_2024-01-31.png (SHA-256 42d0bd19b5933eae...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [367, 255], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 475f1a8005785490...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

83 FITS frames (55 MB), HATP-36240131091517.FITS → HATP-36240131132115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-240131.log (f51d90951cef...), FinalLightCurve_HAT-P-36 b_2024-01-31.png (42d0bd19b593...), FinalLightCurve_HAT-P-36 b_2024-01-31.csv (37ebd664654d...)

Compute resources

Wall time 245.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 03:26Z.